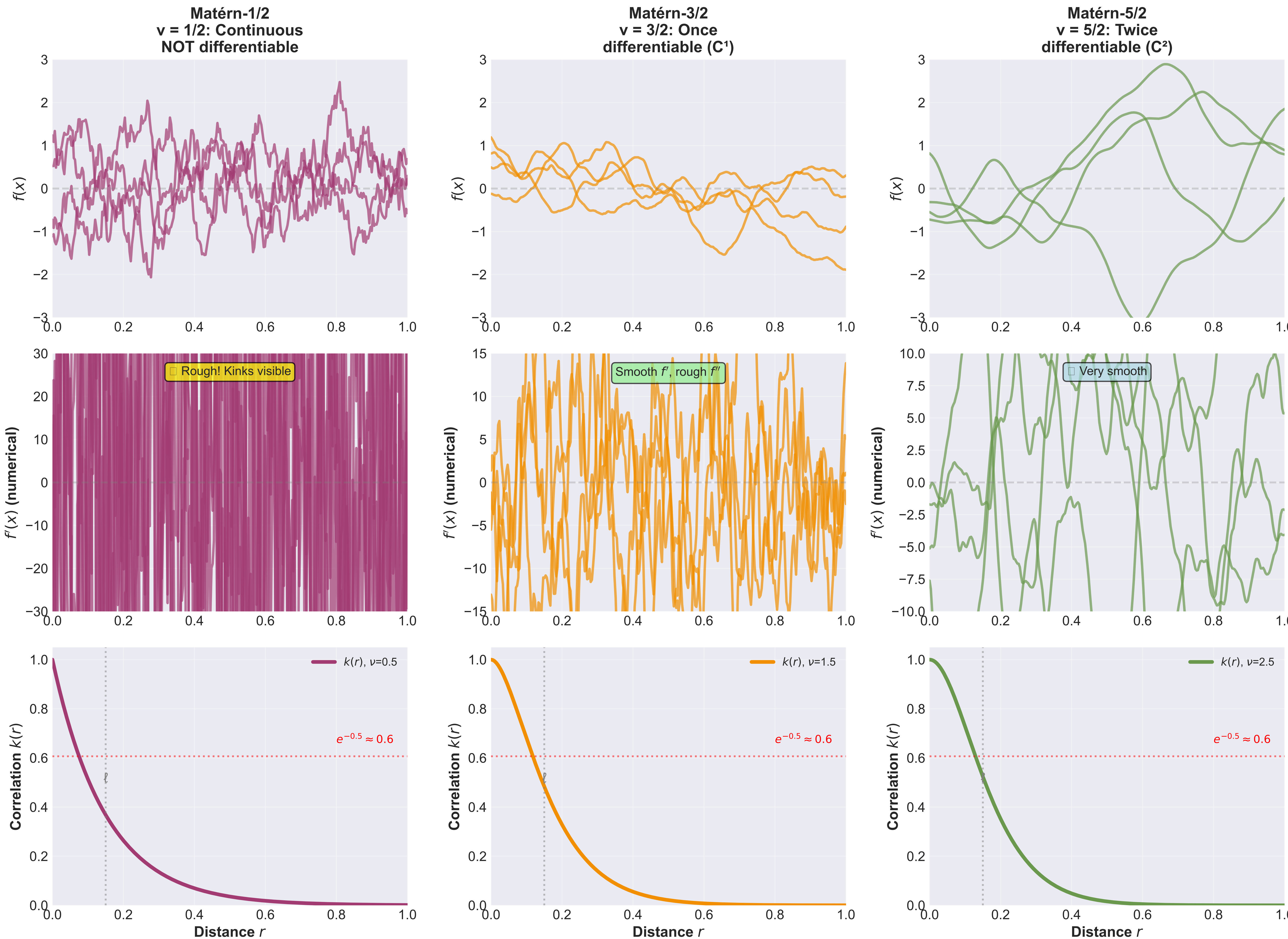
Matérn Smoothness Comparison: How ν Controls Differentiability



Top row: Function samples $f(x)$. Middle row: Numerical derivatives $f'(x)$ reveal roughness—note how ν=1/2 shows kinks (not differentiable), while ν=5/2 is very smooth. Bottom row: Kernel correlation $k(r)$ shows how quickly correlations decay. Matérn-5/2 is recommended default: smooth enough for most physics, more realistic than infinitely-smooth SE.